UG (NEP) 6th SEMESTER
COMPUTER APPLICATIONS – PROBABILITY AND STATISTICS (CAP622J3)

No. of Questions: 09 Session: June 2026
Time Allowed: 2:00 Hours Max. Marks: 100 Min. Marks: 36

NOTE: Attempt all questions from Section “A” and “B” and only two questions from Section “C”.

Section A: Short Answer Type Questions (8 × 2 = 16 Marks)

Answer all questions in a maximum of 20 words each.

i. Define event in probability.
ii. Define addition theorem.
iii. Define covariance.
iv. What is geometric distribution?
v. What is regression line?
vi. What is population mean?
vii. Define simple linear regression.
viii. What is critical region?

Section B: Medium Answer Type Questions (4 × 10 = 40 Marks)

Answer all the four questions with internal choice in a maximum of 150 words each.

2. Discuss simple and compound probability with suitable examples.

OR

Discuss mean, median, and mode. Calculate mean, median, and mode for the data: 4, 6, 8, 6, 5, 9, 6.

3. Explain random variables and probability distributions.

OR

State and explain central limit theorem.

4. Explain residuals and non-linear models.

OR

A sample of 100 students shows 60 students prefer tea. Construct a 95% confidence interval for the population proportion.

5. Discuss simple linear regression and least square principle.

OR

Calculate chi-square value for observed frequencies: 20, 25, 30, 25 and expected frequencies: 25, 25, 25, 25.

Section C: Long Answer Type Questions (2 × 22 = 44 Marks)

Answer only two questions in a maximum of 400 words each.

6. State and explain addition and multiplication theorems of probability. Solve suitable numerical problems based on them.
7. Explain variance and standard deviation. Find the variance and standard deviation of the data: 2, 4, 6, 8, 10.
8. Discuss regression analysis, prediction, and error in prediction.
9. Explain hypothesis testing and tests of significance in detail.